

EXAM – CHALLENGE 1: Exploring the Impact of Natural Environments on Birth Outcomes: A Research Proposal Exercise

Imagine you are a researcher contributing to a study investigating how exposure to natural environments, such as green and blue spaces, and characteristics of the mothers influence birth outcomes. Your specific task is to develop a mini-research project that builds on this theme, focusing on maternal exposure to natural environments and its potential impact on fetal growth and birth weight. The two outcomes that need to be addressed are the birth weight of the newborns (regression task) and the prematurity of the newborns (binary classification task).

Part 1: Introduction

Write a brief overview of your research proposal, addressing the following:

- **Research Question:** Define a clear, focused research question that your study will address.
- **Hypothesis:** Based on prior findings, state your hypothesis. For example, how do you expect maternal exposure to green and blue spaces to affect birth weight?
- **Rationale:** Explain why this research question is important. Refer to findings from the literature studies attached in the folder, particularly the observed benefits of green space exposure and the limited data on blue spaces.

Part 2: Materials and Methods

Develop a basic outline for the research methods you would use, considering the following components:

- **Study Population and Sample:** Describe the characteristics of your study participants (e.g., number of participants and any special considerations such as socioeconomic status).
- **Covariates:** List relevant covariates you would control for in your analysis (e.g., air pollution levels, indicators). Describe why these are important for interpreting results accurately.
- **Outcome Measurement:** Describe the main outcomes you would examine (e.g., birth weight).

Describe the statistical approach you would use to analyze your data:

- **Statistical Models:** Specify the types of regression models (e.g., linear, logistic) you would employ to analyze the relationships between natural environment exposure and birth outcomes.

Part 3: Results

In the Results section of your report, summarize the key findings based on the dataset provided.

Describe any associations you found between explanatory variables and the responses for the two tasks.

For example, highlight if greenspace exposure is generally linked with increased birth weight or lower risk of preterm birth, or if blue space has less impact on these outcomes.

Summarize the potential implications of these results, particularly in terms of the benefits of greenspace for mothers from lower socioeconomic backgrounds.

Your results should be presented clearly and concisely, focusing on observed patterns in the data without making conclusions or policy recommendations, which will be addressed in the Discussion section.

Part 4: Discussion and Conclusions

Write a brief discussion of your research findings and how they might contribute to public health or urban planning policy.

Grading Criteria:

1. Clarity and focus of research question and hypothesis
2. Coherence between python code and the methodology described in the paper
3. Relevance and depth of methodology
4. Appropriateness and rigor of statistical analysis approach
5. Insightfulness of discussion and policy recommendations

Dataset Variable Descriptions:

This dataset includes a range of maternal, environmental, and birth outcome variables relevant to examining how exposure to natural environments might affect birth outcomes. Below is an overview of each variable:

Maternal and Socioeconomic Variables

Maternal_age: Continuous variable representing the mother's age in years at the time of birth.

Parity_Multipara: Binary variable indicating whether the mother has had previous births (1 = multipara, 0 = primipara).

Maternal_education: Categorical variables representing the mother's highest level of education, with "High" indicating post-secondary education, "Medium" indicating secondary education and "Low" primary education.

Maternal_ethnicity: Binary variable indicating whether the mother's ethnicity is Western or not.

Area_level_deprivation: Categorical variables representing the level of area-level deprivation in the mother's residential area.

Maternal_smoking: Binary variable indicating if the mother smoked during pregnancy.

Maternal_alcohol_consumption: Binary variable indicating if the mother consumed alcohol during pregnancy.

Environmental Exposure Variables

NDVI_500m, _300m, _100m: Continuous variable representing the average Normalised Difference Vegetation Index (NDVI) within a 500/300/100-meter radius of the mother's residence, measuring the surrounding greenspace density.

Residential_distance_to_green_space: Continuous variable indicating the distance in meters from the mother's residence to the nearest green space.

Accessibility_to_green_space: Binary variable indicating if there is accessible green space near the mother's residence.

Residential_distance_to_blue_space: Continuous variable indicating the distance in meters from the mother's residence to the nearest blue space (e.g., lake, river, sea).

Accessibility_to_blue_space: Binary variable indicating if there is accessible blue space near the mother's residence.

Air Pollution Variables

NO2: Continuous variable representing the level of nitrogen dioxide (NO₂) pollution in the mother's residential area, measured in micrograms per cubic meter (µg/m³).

PM2.5: Continuous variable representing the level of particulate matter (PM_{2.5}) pollution in the mother's residential area, measured in micrograms per cubic meter (µg/m³).

Birth Outcome Variables

Birth_weight: Continuous variable indicating the birth weight of the newborn in grams.

Pre_term: Binary variable indicating whether the birth was preterm (1 = preterm, 0 = term).

Avoid using *Birth_weight* as input of the model for the *Pre_term* outcome and viceversa.

Cohort Variable

Cohort: Categorical variable indicating the specific cohort or study group to which each participant belongs, allowing for control of regional or cohort-specific effects.

Deadline: 3 December 2024 11:00 CET